

Each *Action description* is authorized by a signature, called the *spend authorization signature*.

Let $\ell_{\text{Merkle}}^{\text{Orchard}}$ be as defined in § 5.3 ‘*Constants*’ on p. 77.

Let $q_{\mathbb{P}}$ be as defined in § 5.4.9.6 ‘*Pallas and Vesta*’ on p. 108.

Let $\text{Extract}_{\mathbb{P}}$ be as defined in § 5.4.9.7 ‘*Coordinate Extractor for Pallas*’ on p. 109.

Let $\text{ValueCommit}^{\text{Orchard}}.\text{Output}$ be as defined in § 4.1.8 ‘*Commitment*’ on p. 32.

Let $\text{SpendAuthSig}^{\text{Orchard}}$ be as defined in § 4.15 ‘*Spend Authorization Signature (Sapling and Orchard)*’ on p. 59.

Let $\text{KA}^{\text{Orchard}}$ be as defined in § 4.1.5 ‘*Key Agreement*’ on p. 28.

Let Sym be as defined in § 4.1.4 ‘*Symmetric Encryption*’ on p. 28.

Let ZKAction be as defined in § 4.1.13 ‘*Zero-Knowledge Proving System*’ on p. 36.

Let vk be the Halo 2 verifying key identified by the Orchard circuit version, as specified in the consensus rules below.

An *Action description* comprises $(\text{cv}^{\text{net}}, \text{rt}^{\text{OrchardPool}}, \text{nf}, \text{rk}, \text{spendAuthSig}, \text{cm}_x, \text{epk}, \text{C}^{\text{enc}}, \text{C}^{\text{out}}, \text{enableSpends}, \text{enableOutputs}, \pi)$ where:

- $\text{cv}^{\text{net}} : \text{ValueCommit}^{\text{Orchard}}.\text{Output}$ is the *value commitment* to the value of the input *note* minus the value of the output *note*;
- $\text{rt}^{\text{OrchardPool}} : \{0 \dots q_{\mathbb{P}} - 1\}$ is an *anchor*, as defined in § 3.4 ‘*Transactions and Treestates*’ on p. 19, for the output *treestate* of a previous *block*;
- $\text{nf} : \{0 \dots q_{\mathbb{P}} - 1\}$ is the *nullifier* for the input *note*;
- $\text{rk} : \text{SpendAuthSig}^{\text{Orchard}}.\text{Public}$ is a randomized *validating key* that should be used to validate spendAuthSig ;
- $\text{spendAuthSig} : \text{SpendAuthSig}^{\text{Orchard}}.\text{Signature}$ is a *spend authorization signature*, validated as specified in § 4.15 ‘*Spend Authorization Signature (Sapling and Orchard)*’ on p. 59;
- $\text{cm}_x : \{0 \dots q_{\mathbb{P}} - 1\}$ is the result of applying $\text{Extract}_{\mathbb{P}}$ to the *note commitment* for the output *note*;
- $\text{epk} : \text{KA}^{\text{Orchard}}.\text{Public}$ is a *key agreement public key*, used to derive the key for encryption of the *transmitted note ciphertext* (§ 4.20 ‘*In-band secret distribution (Sapling and Orchard)*’ on p. 70);
- $\text{C}^{\text{enc}} : \text{Sym}.\text{C}$ is a ciphertext component for the encrypted output *note*;
- $\text{C}^{\text{out}} : \text{Sym}.\text{C}$ is a ciphertext component that allows the holder of the *outgoing cipher key* (which can be derived from a *full viewing key*) to recover the recipient *diversified transmission key* pk_d and the *ephemeral private key* esk , hence the entire *note plaintext*;
- $\text{enableSpends} : \mathbb{B}$ is a flag that is set in order to enable non-zero-valued spends in this *Action*;
- $\text{enableOutputs} : \mathbb{B}$ is a flag that is set in order to enable non-zero-valued outputs in this *Action*;
- $\text{enableCrossAddress} : \mathbb{B}$ is a flag that is set in order to enable the *expanded receiver* of the output *note* to be different from the *expanded receiver* of the input *note* in this *Action* [ZIP-2006];
- $\pi^{\text{OrchardPool}} : \text{ZKAction}.\text{Proof}$ is a *zk-SNARK proof* for the *Action statement* defined in § 4.18.4 on p. 66, with *primary input* $(\text{cv}, \text{rt}^{\text{OrchardPool}}, \text{nf}, \text{rk}, \text{cm}_x, \text{enableSpends}, \text{enableOutputs}, \text{disableCrossAddress})$ where $\text{disableCrossAddress} = 1 - \text{enableCrossAddress}$.

Note: The $\text{rt}^{\text{OrchardPool}}$, $\text{enableSpendsOrchard}$, and $\text{enableOutputsOrchard}$ components are the same for all *Action transfers* in a *transaction*, and are encoded once in the *transaction body* (§ 7.1 ‘*Transaction Encoding and Consensus*’ on p. 125), not the *ActionDescription* structure. $\pi^{\text{OrchardPool}}$ is aggregated with other *OrchardPoolAction* proofs and encoded in the proofsOrchard field.